

AttentionChoice: Discrete choice modelling supported by a deep learning attention mechanism

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Abstract

Discrete choice modelling is a commonly used approach for transport planning based on a Random Utility Maximisation theory. It provides a robust econometric framework to model people's choice behaviours. The advantage of discrete choice models is their interpretability through meaningful coefficients in utility functions. The utility functions represent modellers' assumptions on the relationship between alternative attributes and various personal characteristics. Since the relationships are often non-linear, it is difficult to find a desirable choice model specification that can account for various personal tastes. This requires choice modellers to specify and check various utility functions manually. Advanced machine learning can help to address this issue by providing more flexibility in choice model specifications. This paper provides a perspective on how an attention mechanism in deep learning can be used to capture systematic taste heterogeneity in discrete choice models. To this end, our framework employs parameters in the utility functions as the interaction of personal characteristics with the attention mechanism. Experimental results from a stated preference survey and a revealed preference survey show that our proposed model not only maintains the interpretability of discrete choice models, but also significantly improves its predictive capability.

Keywords: Discrete choice model; Interpretability; Mode choice; Attention mechanism; Deep learning.

1 Introduction

Random Utility Maximisation theory by [McFadden \(1974\)](#) provides a powerful econometric framework to model people's choice behaviours ([Van Cranenburgh et al., 2022](#)). Discrete choice models are a widespread approach to analyse travel mode choice and destination choice in transportation planning. In the choice model specification, modellers are required to specify utility functions which represent their assumptions about the relationships between alternative attributes and various personal characteristics. Each utility function consists of a systematic utility part with a random variable. The systematic component is used to describe how a choice-maker values each alternative attribute (taste), and how tastes vary across different choice-makers (taste heterogeneity ([Han et al., 2020](#))). For example, in mode choice, the commonly used attributes are travel time and travel cost of different travel modes such as car and public transport. The parameters related to travel time and cost in utility functions can represent the sensitivity of travellers to the time and cost of their trips.

In practice, it is not easy to form desirable choice model specifications ([Ben-Akiva and Lerman, 1985](#)). Based on prior knowledge and assumptions, choice modellers need to check various utility functions in order to find a model specification that can reasonably represent the relationship among

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alternative attributes and the characteristics of choice-makers. In the case of mode choice, analysts might be interested in modelling the time and cost sensitivity among different demographic groups of different genders or ages. Choice modellers typically represent the interactive relationship by a product of different alternative attributes (like travel time and cost) and personal features (like gender and age). However, the task can become complicated when including more personal characteristics such as personal income, age, occupation, and employment status. In this case, the number of model specifications can become exponentially large. In addition, when the underlying relationships are non-linear, determining an accurate utility specification can be difficult (Han et al., 2020).

It is noted that finding the “most appropriate” choice model specification is not a simple task. The art of model building requires both statistical methods and modellers intuitive judgements (Ben-Akiva and Lerman, 1985). With prior assumptions, there is a large number of model specifications. The test of many possible different specifications can also be labour-intensive (Van Cranenburgh et al., 2022).

Data-driven machine learning approaches are beneficial for choice model building, selection and evaluation (Van Cranenburgh et al., 2022). These methods can help overcome limitations of choice models in search of the optimal model specification. Recent studies show promising results in incorporating neural networks into discrete choice models to capture taste heterogeneity (Van Cranenburgh et al., 2022). For instance, Van Cranenburgh (2020) uses convolutional neural networks (CNN) to represent heterogeneous tastes, while Wong and Farooq (2021) leverage residual neural networks (ResNet) to account for unobserved heterogeneity in the choice utility function. In addition, systematic taste variety can be captured by using different deep neural network architectures (Siffringer et al., 2020; Wang et al., 2020; Han et al., 2020). These studies show that it is possible to maintain choice model interpretability and improve its predictive capability.

Unlike previous research incorporating deep learning into choice models, this study provides a new perspective of how the attention mechanism can help account for systematic taste heterogeneity. The attention mechanism is a concept derived from attention cues in biology, where people pay attention to a certain thing in a specific context (Zhang et al., 2021). This conceptualisation is encoded by using an attention function, which includes three components: query, key, and value. The use of the attention mechanism in a transformer architecture has recently led to the breakthrough in deep learning research (Vaswani et al., 2017). The transformer models have been applied in various machine learning fields such as natural language processing and computer vision, showing significant performance improvements compared to traditional neural network architectures (Devlin et al., 2018; Carion et al., 2020; Zhang et al., 2021). In this study, we investigate how the attention mechanism can help to explain taste heterogeneity in discrete choice models. We assume that people’s tastes depend on their attention in the decision-making context. To our knowledge, this is the first paper incorporating the attention mechanism to capture systematic taste heterogeneity for choice models. Our contributions include:

- Proposing an AttentionChoice framework that incorporates attention mechanism to better explain taste heterogeneity.
- Maintaining the interpretability, while significantly improving the predictive capability for choice models.
- Helping choice modellers to alleviate the adhoc process in finding good specifications for discrete choice models.

The paper is organized as follows: Section 2 briefly presents the existing approaches that integrate neural networks into discrete choice models to capture heterogeneity. In addition, we also introduce

the concept of attention mechanism and how it is used in deep learning research. Next, Section 3 proposes an AttentionChoice framework. After that, the data and model specification used for this study is described in Section 4. Section 5 then presents and discusses the results of the proposed model. Finally, Section 6 summarizes the main conclusions with potential extensions of this research.

2 Background

This section provides a brief overview of recent studies incorporating neural networks into discrete choice models. For a comprehensive review on the use of machine learning, especially deep learning with discrete choice models, readers can refer to [Van Cranenburgh et al. \(2022\)](#).

2.1 Neural networks for discrete choice

There is an increasing trend that applies advanced machine learning, especially deep neural networks in the choice modelling field. While many studies focus on making comparisons in terms of model fit, there is research that goes beyond comparison, and tries to integrate deep learning models into discrete choice models ([Van Cranenburgh et al., 2022](#)). The goal of integration is to get the best of both worlds which consists of model interpretability and predictive capability. Especially, recent advances in discrete choice models with deep neural networks have shown promising results ([Sifringer et al., 2020](#); [Han et al., 2020](#); [Wong and Farooq, 2021](#)). These studies can be beneficial for improved choice modellers by tackling challenges in the search for discrete choice model specifications.

Capturing random heterogeneity

Several studies have developed different theory-driven models to capture random or unobserved heterogeneity, which cannot be associated with observable attributes ([Van Cranenburgh et al., 2022](#)). For example, Latent Class and Mixed Logit models are probably the most popular choice frameworks used to model unobserved heterogeneity. In term of using neural networks to capture unobserved heterogeneity, [Van Cranenburgh \(2020\)](#) combine a flexible convolutional neural network (CNN) with the behavioural notion to represent unobserved taste heterogeneity.

Recently, [Wong and Farooq \(2021\)](#) introduces ResLogit, which is another approach to embed neural networks into a logit model. They leverage a residual neural network (ResNet ([He et al., 2016](#))) to account for unobserved choice heterogeneity in the choice utility function. A residual neural network is the combination of a feed-forward neural network with an identity shortcut connection ([He et al., 2016](#)).

In a Multinomial Logit (MNL) model, $\mathbf{U}_{in}$ is the utility of a person n selecting choice i . $\mathbf{U}_{in}$ consists of a deterministic component $\mathbf{V}_{in}$ and a random error component ϵ_{in} : $\mathbf{U}_{in} = \mathbf{V}_{in} + \epsilon_{in}$. On the other hand, in the ResLogit model, the utility function is defined as:

$$\mathbf{U}_{in} = \mathbf{V}_{in} + \mathbf{g}_n + \epsilon_{in} \quad (1)$$

where $\mathbf{g}_n$ represents the unobserved heterogeneity part, which is calculated as:

$$\mathbf{g}_n^m = \sum_{m=1} \ln(1 + \exp(\theta^m \times \mathbf{h}_n^{(m-1)})) \quad (2)$$

$$\mathbf{h}_n^m = \mathbf{h}_n^{m-1} + \mathbf{g}_n^{m-1} \quad (3)$$

$$\mathbf{h}_n^0 = \mathbf{V}_n \quad (4)$$

where m is the number of residual layers. Thus, the utility function $\mathbf{V}_{in}$ goes through several residual neural networks before input into a logit function to calculate the final choice (see Figure 1 (b)).

They show that ResLogit is able to improve the representation of non-linearity and cross-effects in choice utilities (Wong and Farooq, 2021). The estimation of parameters using data-driven approach can help to address model identifiability and yield smaller standard errors than the MNL model. Furthermore, the use of ResLogit method outperform traditional multi-layer neural network models in prediction accuracy while providing similar interpretability as the MNL model (Wong and Farooq, 2021). However, ResLogit still cannot address the complexity issue in choice model specifications.

It is worth noting that increasing the model complexity by adding more deep neural network layers may not help to improve model performance. The number of layers in an optimal model does not have an asymptotic limit (He et al., 2016). Studies comparing deep neural networks with discrete choice models also affirm the limitation of putting multiple layers to enhance modelling accuracy (Alwosheel et al., 2018; Lee et al., 2018).

Capturing systematic heterogeneity

Deep learning models can also help to capture systematic heterogeneity such as the various tastes among decision-makers. In the choice model specification, this systematic heterogeneity is usually specified as the *interactions* among variables (covariates) in the utility functions (Van Cranenburgh et al., 2022). However, when the number of features increases, the number of higher-order interactions will grow exponentially, making it difficult if not impossible to manually compare among different utility specifications. This issue can be addressed by using neural networks that does not require analysts to impose explicit prior assumptions (Van Cranenburgh et al., 2022). The embedding of neural networks into discrete choice models can also enhance the model's flexibility to present systematic taste heterogeneity. These proposed models often consist of a part that is transparent and interpretable through an MNL part, while another part which is flexible to capture the interactions (Sifringer et al., 2020; Wang et al., 2020; Han et al., 2020).

For example, Han et al. (2020) proposes a TasteNet model that incorporates discrete choice with neural networks to improve the flexibility in representing taste heterogeneity. In TasteNet, the parameters of travel mode's attributes like travel time and travel cost are learned from personal characteristics using a neural network.

In particular, the authors separate the systematic utility $\mathbf{V}_{in}$ into a parametric part and a flexible part as follows:

$$\mathbf{V}_{in} = \beta_i^{TN} \mathbf{x}_{in}^{TN} + \beta_i^{MNL} \mathbf{X}_{in}^{MNL} \quad (5)$$

where systematic taste heterogeneity is captured by using TasteNet (a feed-forward neural network) to represent taste parameters as functions of individual characteristics (Han et al., 2020):

$$\beta_n^{TN} = \text{TasteNet}(\mathbf{Z}_n; \theta) \quad (6)$$

and $\mathbf{Z}_n$ is the characteristics of the choice-maker n , and θ are unknown parameters of a neural network.

This approach has improved the model's predictability while maintaining its interpretability. They show that the TasteNet can significantly improve the prediction accuracy of the MNL model (Han

et al., 2020). However, TasteNet only uses a traditional feed-forward neural network (also named as a multi-layer perceptron (MLP)). The MLP architecture suffers from non-identifiability because of the accumulation of non-linear activation functions in several layers (Wong and Farooq, 2021). Repeated non-linear transformations across layers lead to model degradation when increasing the number of layers (He et al., 2016). In addition, the performance of TasteNet does not improve with more hidden feed-forward layers (Han et al., 2020).

In summary, Fig. 1 shows a simplified graphical diagram of recent state-of-the-art models compared with the traditional MNL model. Herein, (Fig. 1a) is the traditional Multinomial Logit Model (MNL). In the middle, (Fig. 1b) is Reslogit model, which formulates the utility function as the product of $\mathbf{X}_{nt}$ and parameters β . Thus, its utility function is similar to the utility function of the MNL model. The difference compared to MNL is that the utility function of ResLogit can go through several residual network layers, before passing on to the last logit layer. On the right, (Fig. 1c) is TasteNet, which uses a neural network h_1 to learn the parameters β from individual characteristics ($\mathbf{Z}_{nt}$ is the characteristics of individual n with choice task t). The utility function V_{nt} is calculated based on the learned parameters beta and the attributes of alternatives ($\mathbf{X}_{nt}$). For example, in mode choice, $\mathbf{X}_{nt}$ can be the travel time and cost of different travel modes. The utility function is then evaluated by using a logit layer to simulate the probability of choosing mode alternatives.

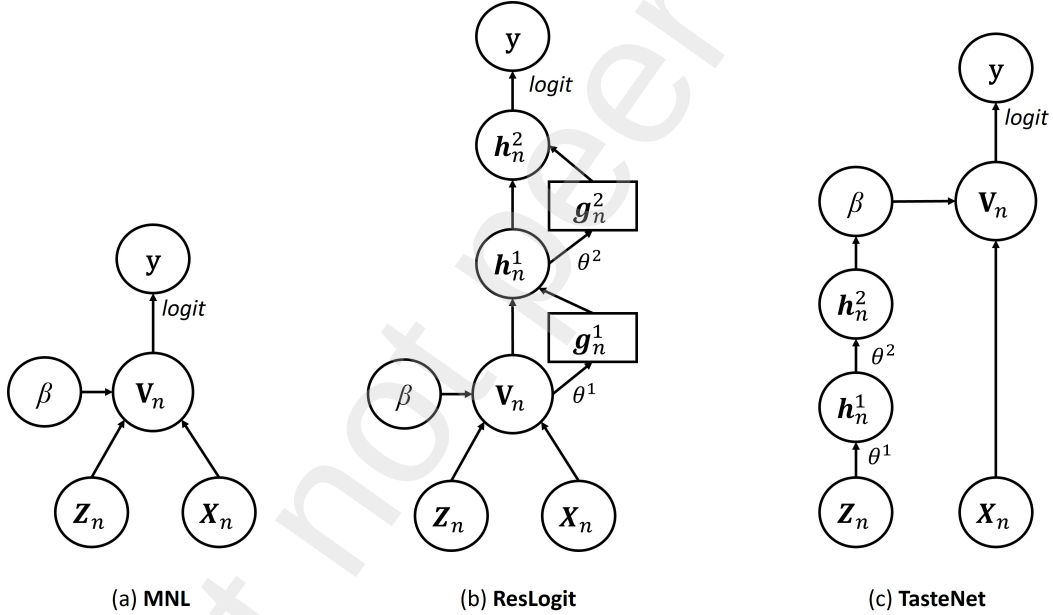


Figure 1: Simplified graphical models: (a) A Multinomial Logit Model; (b) A ResLogit model with two residual layers; (c) A TasteNet model with two feed-forward hidden layers.

Different to previous studies, in this paper, we propose a new framework to incorporate an attention mechanism to account for taste heterogeneity in discrete choice models.

2.2 Attention mechanism

In deep learning, an attention mechanism is used to mimic human cognitive attention. The idea stems from attention cues in biology, where human attention is a limited and valuable resource (Zhang et al., 2021). For example, when a person looks at a specific object and focuses her attention on that object, other objects are often diminished. James (2007) introduced a framework to explain how an attention mechanism is deployed by using both the nonvolitional cue and volitional cue (Zhang et al., 2021). The nonvolitional cue is based on the saliency of objects in the environment. Besides, the volitional cue, based on variable selection criteria under cognitive control, is deliberate attention

(Zhang et al., 2021).

Inspired by the nonvolitional and volitional attention cues, researchers have designed an attention mechanism by using queries, keys and values with attention pooling in machine learning architectures. Attention pooling is designed so that the given queries (volitional cue) can interact with keys (nonvolitional cues), which guides bias selection over values (input data) (Zhang et al., 2021). The effect of the attention mechanism can enhance some parts of the input information while ignoring other parts. The motivation here is that the neural network should devote more resources (higher weights) to the small, but important, parts of the inputs.

Attention functions

An attention function can be considered as mapping a query ($\mathbf{q}$) with a set of key ($\mathbf{k}$) and value ($\mathbf{v}$) pairs to an output. The output is computed as a weighted sum of the values, where the weight assigned to each value is computed by a compatibility function of the query with the corresponding key (Vaswani et al., 2017). The two most used attention functions are additive attention and multiplicative (dot-product) attention. The dot-product attention has a more computationally efficient design, which leads to faster and more space-efficient in implementation (Vaswani et al., 2017; Zhang et al., 2021).

The scaled dot-product attention function is defined as:

$$\text{Attention}(\mathbf{q}, \mathbf{k}, \mathbf{v}) = \text{softmax}\left(\frac{\mathbf{q}\mathbf{k}^\top}{\sqrt{d}}\right) \mathbf{v} \quad (7)$$

where ($\mathbf{q}$) is a query, ($\mathbf{k}$) and ($\mathbf{v}$) are a set of key and value pairs respectively, $\text{softmax}(i) = \frac{e^i}{\sum_{j \in D} e^j}$. It is noted that softmax (in deep learning) and logit function (in discrete choice) are mathematically equivalent.

The scaled dot product of both vectors has zero mean and a variance of d , so to ensure that the variance of the scaled dot product remains one regardless of vector length, the dot product is divided by $\sqrt{d}$.

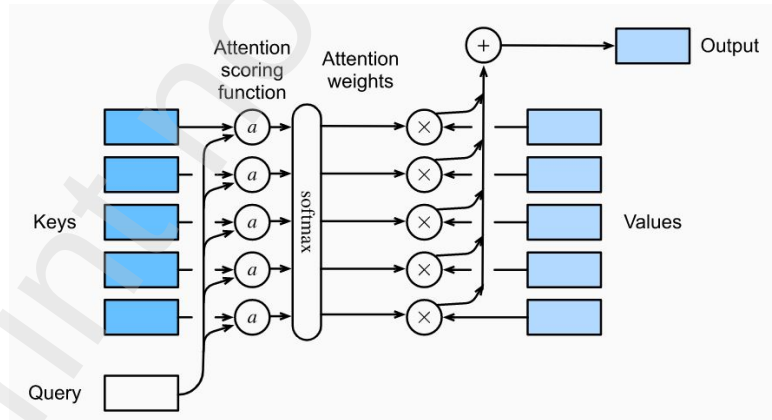


Figure 2: The output of attention pooling is calculated as a weighted average of values (Zhang et al., 2021).

Multi-Head Attention

In practice, given the same set of queries, keys, and values, analysts may want to combine knowledge from different behaviours of the same attention mechanism, such as capturing dependencies of various travel time ranges (e.g., shorter-range vs. longer-range) within a sequence. Hence, instead of using a single attention function, it could be beneficial to allow the attention mechanism to jointly use different representation subspaces of queries, keys, and values (Vaswani et al., 2017).

Multi-head attention (MHA) allows the model to jointly attend to information from different representation subspaces at different positions. In the end, attention pooling outputs are concatenated and transformed with another learned linear layer h to produce the final output (Vaswani et al., 2017).

Mathematically, given a query $\mathbf{q}$, a key $\mathbf{k}$, and a value $\mathbf{v}$, each attention head is computed as

$$\text{Attention} = f(\theta^q \mathbf{q}, \theta^k \mathbf{k}, \theta^v \mathbf{v}) \quad (8)$$

where learnable parameters θ^q , θ^k , and θ^v , and f is the scaled dot-product attention in Eq.7. The multi-head attention output is another linear transformation via learnable parameters θ_o of the concatenation of m attention heads:

$$\text{MHA} = \theta^o \times \text{Concatenate}(\text{Attention}_1, \text{Attention}_2, \dots, \text{Attention}_m) \quad (9)$$

Based on this design, each head may attend to different parts of the input (Vaswani et al., 2017). Thus, more sophisticated functions than the simple weighted average can be expressed. This can help to yield more interpretable models by learning different tasks simultaneously (Vaswani et al., 2017). For a more detailed explanation of the attention mechanism, readers can refer to Rush (2018).

Using Multi-head attention layers in deep learning architectures has led to breakthroughs in deep learning research. Numerous studies extending the transformer model have shown its efficacy in different fields such as natural language processing and computer vision (Devlin et al., 2018; Carion et al., 2020). In the next section, we present a framework that incorporates a multi-head attention layer into the discrete model to capture taste heterogeneity in utility functions.

3 AttentionChoice model

3.1 Model architecture

In this section, we propose a framework that integrates the attention mechanism into a discrete choice model. Fig. 3 depicts the proposed AttentionChoice framework using an attention mechanism. In this framework, the parameter β is learned from individual characteristics $\mathbf{Z}$ by using a multi-head attention (MHA) layer with two attention heads.

Here, the systematic utility function $\mathbf{V}$ of person n is defined as:

$$\mathbf{V}_{in} = \beta_i^{\text{MHA}} \mathbf{X}_{in}^{\text{MHA}} + \beta_i^{\text{MNL}} \mathbf{X}_{in}^{\text{MNL}} \quad (10)$$

where systematic taste heterogeneity is captured by using a *MultiHeadAttention* layer:

$$\beta_n^{\text{MHA}} = \text{MultiHeadAttention}(\mathbf{Z}_n; \theta) \quad (11)$$

This leads to the choice probability $\mathbf{P}_{in}$ for $\forall i \in (1, \dots, J)$:

$$\mathbf{P}_{in} = \mathbf{P}(\mathbf{y}_n = i) = \frac{\exp(\mathbf{V}_{in})}{\sum_{j \in (1, \dots, J)} \exp(\mathbf{V}_{jn})} \quad (12)$$

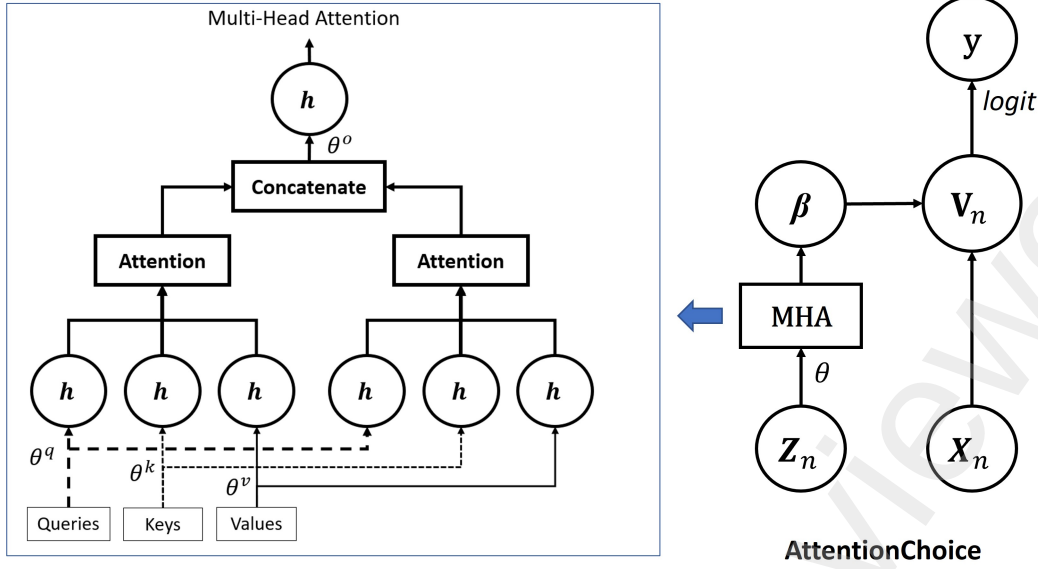


Figure 3: The proposed AttentionChoice model with $\beta = \beta_n^{\text{MHA}} = \text{MultiHeadAttention}(\mathbf{z}_n; \theta)$

We expect that the above attention mechanism can capture and explain the systematic taste heterogeneity. It is because, for instance, in the case of travel mode choice, the β parameters can be the travel time and cost of different modes such as car and train. People with different characteristics can have different tastes or sensitivity to the travel time and cost among different alternatives. The tastes could also be dependent on the individual attention and context. For example, a student with a limited travel budget could pay more attention to the travel cost to the university. However, after graduation, the same person may be more concerned with travel time than cost while travelling to work. Thus, the attention mechanism can learn the taste coefficients β according to various personal demographic information such as age, employment status, and income.

3.2 Model estimation

The stochastic gradient descent (SGD) algorithm is used for the estimation procedure. We evaluate the predictive performance on an out-of-sample validation set. Estimating a large number of parameters using a pure optimization method can lead to non-identifiability issues (Wong and Farooq, 2021). Hence, we estimate these models by using the data-driven SGD approach to avoid the non-identifiability issues.

AttentionChoice model is estimated by optimising an objective function that maximizes the Log-likelihood or minimize the Negative Log-likelihood (NLL), which is defined as follows:

$$\text{NLL} = - \sum_{n=1}^N \ln P(i_n | \mathbf{X}_n; \theta, \beta) \quad (13)$$

where learnable parameters θ and β are updated by the SGD algorithms in each training iteration (Goodfellow et al., 2016). Here, θ and β are adjusted to reduce training errors, while we keep monitoring validation errors to avoid model overfitting.

3.3 Model evaluation

The *accuracy* (Eq. (14)) and *F1 score* (Eq. (15)) are used to evaluate the predictive performance among models. F1 can be interpreted as a harmonic mean of precision and recall as defined below. In

Table 1: Description of variables in the Swissmetro data

Variable	Description	Data type
Alternative attributes (X)		
TRAIN_TT	Train travel time in minutes	Continuous
TRAIN_CO	Train travel cost	Continuous
TRAIN_HE	Train headway in minutes	Continuous
TRAIN_AV	Train availability	Dummy
SM_TT	Swissmetro travel time in minutes	Continuous
SM_CO	Swissmetro travel cost	Continuous
SM_HE	Swissmetro headway in minutes	Continuous
SM_SEATS	Airline seats	Dummy
SM_AV	Swissmetro availability	Dummy
CAR_TT	Car travel time in minutes	Continuous
CAR_CO	Car travel cost	Continuous
CAR_AV	Car availability	Dummy
Personal features (Z)		
AGE	The age group of individuals: 6 groups	Categorical
MALE	Traveler's gender: female or male	Dummy
INCOME	Traveler's income per year	Categorical
GA	The Swiss annual season ticket	Dummy
PURPOSE	Travel purpose: 8 options	Categorical
LUGGAGE	Number of pieces: 3 options	Categorical

addition, Negative Log-likelihood (NLL), the objective function in Eq. (13), is also calculated.

$$\text{Accuracy}^1 = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{FP} + \text{TN} + \text{FN}} \quad (14)$$

$$\text{F1} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} = \frac{\text{TP}}{\text{TP} + 0.5 \cdot (\text{FP} + \text{FN})} \quad (15)$$

4 Case studies

4.1 Case study 1

Swissmetro dataset

We implement the proposed AttentionChoice model for travel mode choice using a Swissmetro dataset, which was collected by Bierlaire et al. (2001). The Swissmetro dataset is a stated preference survey from 1,192 respondents, each with nine choices (Bierlaire, 2018). The universal choice set consists of Car (CAR), Train (TRAIN), and Swissmetro (SM). The description of different features of alternatives, travellers, and trips is presented in Table 1. Readers can refer to Bierlaire (2018) for more detailed information.

A data pre-processing step is performed where samples with unknown age or unknown choice are removed. After data cleaning, there are 10,692 observations, which then be randomly split into training and evaluation sets with 7,484 and 3,208 observations, respectively. The proportions of chosen modes in the training and validation set are shown in Fig. 4. The SM mode consists of around 58%, followed by CAR at roughly 29% and TRAIN at about 13% in both data sets.

The training data set is used to build models, while the validation set is used to validate how well the estimated parameters in these models can be generalized in out-of-sample data. We use several

¹TP: True Positives, TN: True Negatives, FP: False Positives, FN: False Negatives

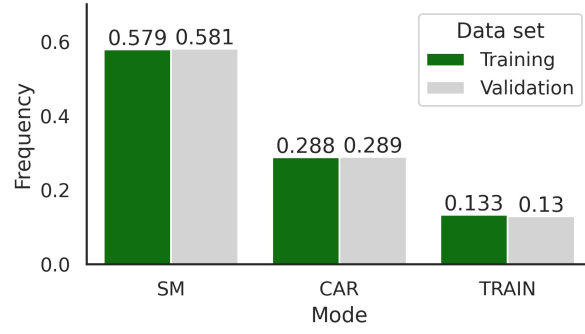


Figure 4: The proportions of different modes in train and validation set

metrics, including Negative Log-likelihood (NLL), *accuracy* (ACC) and F1 score to evaluate model performance (Section 3.3).

Model specification

In the MNL model, the random utility of user n selecting alternative zone i can be written as the sum of a systematic part V and a random error term ϵ (Ben-Akiva and Lerman, 1985). The assumption is that ϵ is an independent identical extreme value distribution. The probability of user n selecting alternative zone i is described as:

$$P_{in} = \frac{e^{\mu V_{in}}}{\sum_{j \in C_n} e^{\mu V_{jn}}} \quad (16)$$

where C_n is the true choice-set of J_n alternatives from which user n chooses an alternative, and the scale μ is usually normalized to one.

In this implementation, it is assumed that all discrete choice coefficients (taste parameters) are represented as the output β^{TN} of TasteNet or the output β^{MHA} of AttentionChoice, so the β^{MNL} is set as empty. Thus, the systematic utility functions of different travel modes in TasteNet and AttentionChoice are defined as:

$$\mathbf{V}_{\text{TRAIN}} = \beta_{\text{train_tt}} * \mathbf{X}_{\text{train_tt}} + \beta_{\text{train_cost}} * \mathbf{X}_{\text{train_cost}} + \beta_{\text{train_hw}} * \mathbf{X}_{\text{train_hw}} \quad (17)$$

$$\mathbf{V}_{\text{SM}} = \text{ASC}_{\text{sm}} + \beta_{\text{sm_tt}} * \mathbf{X}_{\text{sm_tt}} + \beta_{\text{sm_cost}} * \mathbf{X}_{\text{sm_cost}} + \beta_{\text{sm_hw}} * \mathbf{X}_{\text{sm_hw}} \quad (18)$$

$$\mathbf{V}_{\text{CAR}} = \text{ASC}_{\text{car}} + \beta_{\text{car_tt}} * \mathbf{X}_{\text{car_tt}} + \beta_{\text{car_cost}} * \mathbf{X}_{\text{car_cost}} \quad (19)$$

where TasteNet's parameters $\beta^{TN} = \text{TasteNet}(\mathbf{Z}, \theta)$, and AttentionChoice's parameters $\beta^{MHA} = \text{MultiHeasAttention}(\mathbf{Z}, \theta)$, with $\mathbf{Z}$ are personal characteristics including age, gender, income, annual ticket, travel purpose and the number of luggages (as in Table 1).

We also implement MNL and ResLogit models for comparison. To make the results comparable, in addition to the above utility functions, the personal characteristics $\mathbf{Z}$ are also included in the utility functions of MNL and ResLogit along with their specific coefficients.

Table 2: Description of variables in the London Passenger data

Variable	Description	Data type
Alternative attributes (X)		
dur_pt	Train travel time in hours	Continuous
cost_transit	Train travel cost in GBP	Continuous
pt_interchanges	Number of interchanges	Continuous
pt_av	Train availability	Dummy
dur_driving	Car travel time in hours	Continuous
cost_driving	Car travel cost in GBP	Continuous
car_av	Car availability	Dummy
dur_cycling	Cycling travel time in hours	Continuous
cycle_av	Cycle availability	Dummy
dur_walking	Walking time in hours	Continuous
walk_av	Walk availability	Dummy
Personal features (Z)		
age	The age group of individuals: 6 groups	Categorical
female	Gender: female (0), male (1)	Dummy
driving_license	Driving license	Dummy
car_ownership	Car ownership: 3 options	Categorical
purpose	Travel purpose: 5 options	Categorical
fueltype	Fuel type: 6 options	Categorical
faretype	Fare type: 5 options	Categorical
start_time	Start time in hours	Continuous
travel_date	Travel Date: from 1 to 31	Categorical
travel_month	Travel Month: from 1 to 12	Categorical
day_of_week	Day of week: from 1 to 7	Categorical

4.2 Case study 2

London Passenger dataset

The London Passenger dataset is a revealed preference survey of trip diary data of people in London from 2012 to 2015 (Hillel et al., 2018). It includes details for 81,086 trips made by 31,954 individuals. The universal choice set consists of Car (CAR), Public transport (PT), Cycling (CYCLE), and Walking (WALK). The description of different features of alternatives, travellers, and trips is presented in Table 2. Readers can refer to Hillel (2019) for more detailed information.

The dataset is randomly split into (70%) training and (30%) evaluation sets with 56,760 and 24,326 observations, respectively. The proportions of chosen modes in the training and validation set are depicted in Fig. 5. The Car mode consists of around 44%, followed by Public transport at roughly 35%, Walking at around 18% and Cycling at about 3% in both training and validation data sets.

The training data set is used to build models, while the validation set is used to validate how well the estimated parameters in these models can be generalized in out-of-sample data.

Model specification

Similar to the previous case of Swissmetro, in this implementation for London Passenger data, the systematic utility functions of different travel modes in TasteNet and AttentionChoice are defined as:

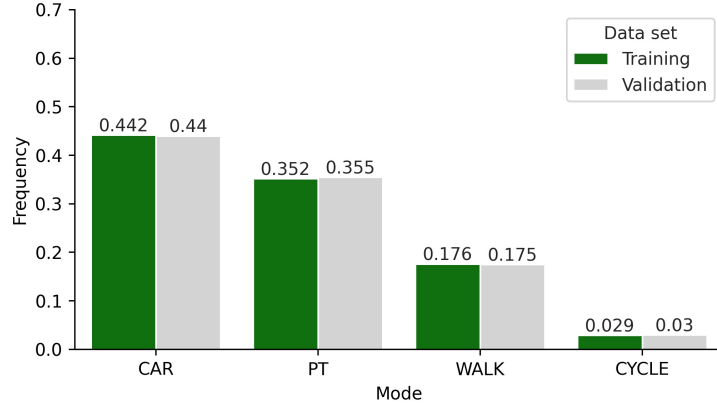


Figure 5: The proportions of different modes in train and validation set

$$\mathbf{V}_{\text{CAR}} = \beta_{\text{cost_driving}} * \mathbf{X}_{\text{cost_driving}} + \beta_{\text{dur_driving}} * \mathbf{X}_{\text{dur_driving}} \quad (20)$$

$$\mathbf{V}_{\text{PT}} = \text{ASC}_{\text{pt}} + \beta_{\text{cost_transit}} * \mathbf{X}_{\text{cost_transit}} + \beta_{\text{dur_pt}} * \mathbf{X}_{\text{dur_pt}} \quad (21)$$

$$\mathbf{V}_{\text{CYCLE}} = \text{ASC}_{\text{cycle}} + \beta_{\text{dur_cycling}} * \mathbf{X}_{\text{dur_cycling}} \quad (22)$$

$$\mathbf{V}_{\text{WALK}} = \text{ASC}_{\text{walk}} + \beta_{\text{dur_walking}} * \mathbf{X}_{\text{dur_walking}} \quad (23)$$

where TasteNet’s parameters $\beta^{TN} = \text{TasteNet}(\mathbf{Z}, \theta)$, and AttentionChoice’s parameters $\beta^{MHA} = \text{MultiHeadsAttention}(\mathbf{Z}, \theta)$, with $\mathbf{Z}$ are personal characteristics including age, gender, car ownership, trip purpose, and other features (as in Table 2). We also implement MNL and ResLogit models for comparison.

5 Results and discussions

We implement the AttentionChoice with one MultiHeadAttention layer including eight attention heads. The TasteNet consists of two feed-forward neural layers, and the ResLogit has eight residual layers. Small learning rates ($1e^{-6}$, $1e^{-5}$, $1e^{-4}$, $1e^{-3}$, $1e^{-2}$) are used for estimating parameters with 5,000 to 30,000 iterations. Note that this study only shows a relative comparison between these models. It is not an exhausting search to find the best models. This experiment aims to show the effectiveness of the neural networks, especially the attention mechanism as a way of representing taste heterogeneity in the utility functions of the MNL model.

5.1 Case study 1: Swissmetro

The estimated parameters

Table 3 compares estimated coefficients across different models. Compared to the traditional MNL model, we observe that the three other models can generate similar coefficient values. The signs of different coefficients in TasteNet and AttentionChoice are similar to that of MNL, while there are differences between TRAIN’s travel cost and SM’s travel cost parameters in ResLogit. Nevertheless, the results show that incorporating different deep learning architectures can maintain the interpretability of discrete choice models.

The detail of estimated parameters along with standard errors of all models, and the cumulative distributions of important alternative parameters are attached in [Appendix A. Swissmetro](#).

Table 3: Comparison of a subset of estimated parameters among different models

Parameter (β)	MNL	ResLogit	TasteNet	AttentionChoice
TRAIN_TT	-1.392	-0.293	-0.078	-0.358
TRAIN_HE	-0.006	-0.001	-0.019	-0.013
TRAIN_CO	-1.052	0.064	-1.210	-1.606
SM_TT	-1.347	0.059	-0.803	-0.842
SM_SEATS	0.140	-0.085	0.076	0.294
SM_HE	-0.008	0.000	-0.006	-0.008
SM_CO	-0.844	0.060	-1.049	-1.483
CAR_TT	-1.130	-0.054	-1.315	-1.822
CAR_CO	-0.743	-0.040	-0.652	-0.924

Table 4: Value of time (VOT) among different models

	MNL	ResLogit	TasteNet	AttentionChoice
TRAIN VOT	1.3238	-4.6132	0.0645	0.2230
SM VOT	1.5952	0.9752	0.7659	0.5679
CAR VOT	1.5219	1.3434	2.0183	1.9722

Value of time

From the travel time and travel cost coefficients in Table 3, we calculate the relative value of time (VOT) for each mode, where $VOT = \beta_{time} / \beta_{cost}$. Table 4 compares the time value of the three travel modes among different models. For CAR, the VOT values are larger than one in all models. This means those selecting CAR prefer travel time than travel cost. However, for TRAIN and SM mode, while MNL produces a SM VOT value at around 1.6, other models deliver VOT values less than 1. One exception is TRAIN VOT of ResLogit with a negative value. This is due to the estimated coefficient of travel cost of TRAIN mode is positive in the ResLogit model (in the previous Table 3).

Model performance on out-of-sample data

Table 5 demonstrates the predictive capability of different models. It can be seen that the AttentionChoice model can achieve the lowest negative log-likelihood, as well as the highest *accuracy* and F1 scores in both training and out-of-sample valuation set. The *accuracy* of AttentionChoice is 72.0%, followed by TasteNet and ResLogit with 70.6% and 68.0%, respectively. In addition, the F1 score of AttentionChoice is 64.6%, followed by TasteNet at 62.3% and MNL at 55.1%. AttentionChoice outperforms other models in terms of the F1 scores in the training and valuation data. Specifically, there are significant improvements in model performance in terms of *accuracy* and F1 compared to the MNL model.

Fig. 6 shows the confusion matrix of the above models in the out-of-sample validation set. It can

Table 5: Predictive performance of different models on Swissmetro data

Model	Train ACC	Valid ACC	Train F1	Valid F1	Train NLL	Valid NLL
MNL	0.658	0.667	0.534	0.551	0.753	0.739
ResLogit	0.673	0.680	0.527	0.542	0.729	0.713
TasteNet	0.726	0.706	0.661	0.623	0.626	0.682
AttentionChoice	0.760	0.720	0.708	0.646	0.570	0.664

Accuracy (ACC); Negative Log-likelihood (NLL), smaller NLL is better.

Table 6: Comparison of a subset of estimated parameters among different models

Parameter	MNL	ResLogit	TasteNet	AttentionChoice
dur_walking	3.967	-0.647	3.442	3.383
pt_interchanges	-0.683	-0.062	-0.150	-0.326
dur_pt	1.928	-0.418	1.522	1.419
cost_transit	1.587	0.074	0.639	0.978
dur_cycling	4.428	0.637	4.033	3.965
dur_driving	-0.228	-0.244	-1.858	-2.229
cost_driving	0.499	-0.223	0.420	0.459

be seen that the integration of deep learning architectures in the logit model does improve the model performance in the validation set. While it seems to be a slight difference in the predictive performance between ResLogit and TasteNet, there is a significant improvement in the AttentionChoice model compared with the MNL model. Besides, AttentionChoice can deliver more accurate predictions in less popular modes like TRAIN and CAR. This explains why AttentionChoice can achieve better F1 scores than other models. Thus, the attention mechanism can help to capture the heterogeneous tastes in un-popular alternatives.

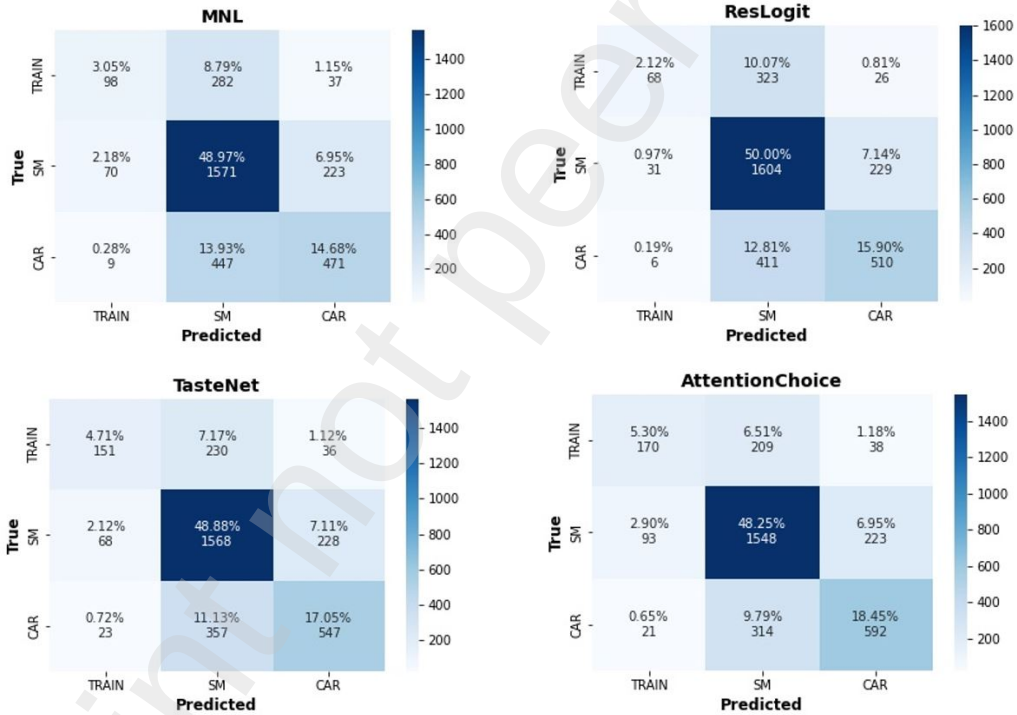


Figure 6: Confusion matrix for Swissmetro data

5.2 Case study 2: London Passenger

The estimated parameters

Table 6 compares estimated coefficients across different models as previously using the London Passenger data. The signs of different coefficients in TasteNet and AttentionChoice are similar to the MNL model, while there are differences from several parameters in ResLogit. The detail of estimated parameters along with standard errors of all models, and the cumulative distributions of important alternative parameters are attached in [Appendix B. London Passenger](#).

Table 7: Value of time (VOT) for PT and CAR in London Passenger data

	MNL	ResLogit	TasteNet	AttentionChoice
PT VOT	1.2147	-5.6771	2.3817	1.4511
CAR VOT	-0.4575	1.0947	-4.4244	-4.8539

Value of time

From the travel duration and travel cost coefficients in Table 6, we calculate the relative value of time (VOT) for car (CAR) and public transport (PT) modes. Table 7 compares the time value of car and public transport modes among different models. The VOT values for public transport in most of the models are larger than one, while the VOT values for car are mostly negative. In addition, the VOT values from TasteNet and AttentionChoice have similar sign with MNL, while ResLogit produces different signs.

Model performance on out-of-sample data

Table 8 shows the predictive capability of different models. Similar to the previous case, the AttentionChoice model can achieve the lowest negative log-likelihood, as well as the highest *accuracy* and F1 scores in both training and out-of-sample valuation sets. The *accuracy* of AttentionChoice is around 70.9%, followed by TasteNet and ResLogit with 70.4% and 70.2%, respectively. In addition, the F1 score of AttentionChoice is 52.9%, followed by TasteNet at 52.4% and ResLogit at 52.2%. AttentionChoice produces better performance compared to other models in terms of the F1 scores in the training and valuation data. Besides, when integrating the attention mechanism into logit choice model, there are significant improvements in model performance in terms of *accuracy* and F1 score compared to the MNL model.

Table 8: Predictive capability of different models on London Passenger data

Model	Train Accuracy	Test Accuracy	Train F1	Test F1	Train NLL	Test NLL
MNL	0.594	0.594	0.455	0.452	0.923	0.926
ResLogit	0.703	0.702	0.523	0.522	0.757	0.762
TasteNet	0.706	0.704	0.527	0.524	0.739	0.752
AttentionChoice	0.715	0.709	0.535	0.529	0.719	0.744

Accuracy (ACC); Negative Log-likelihood (NLL), smaller NLL is better.

Fig. 7 shows the confusion matrix of the above models in the out-of-sample validation set. Observe that there is a significant improvement in the AttentionChoice model compared with the MNL model, especially in the PT and CYCLE modes.

5.3 Discussions

The experimental results from both stated preference (Swissmetro) and revealed preference (London Passenger) datasets show that integrating advanced data-driven deep learning techniques (such as attention mechanism) into discrete choice models can help to improve its performance. The integrations not only reduce the Negative Log-likelihood, but also improve the *accuracy* and F1 score for logit models. Besides, the models can also maintain the interpretability, and help to improve the explanation of taste heterogeneity.

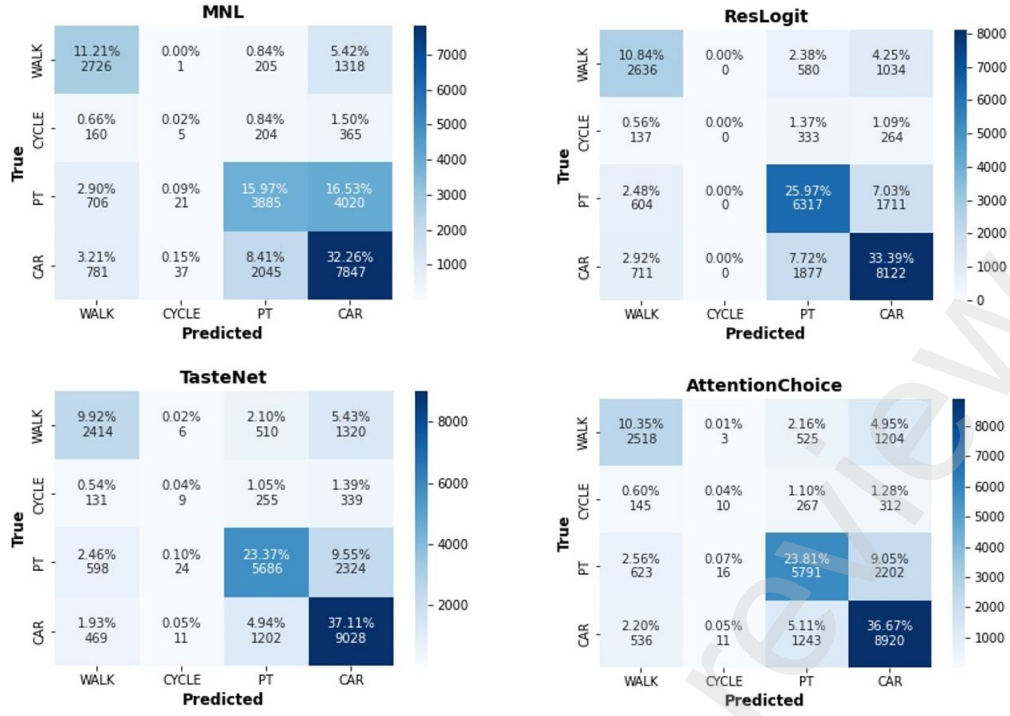


Figure 7: Confusion matrix for London Passenger data

Moreover, the integrated architecture of AttentionChoice can help to reduce the complexity of various choice model specifications. The number of personal characteristics in the Swissmetro and London Passenger datasets is six and eleven, respectively. These numbers could lead to many options in model specifications. Using taste parameters can help alleviate these problems and save the modellers' time. However, modellers still play a vital role in the decision of model specification. For example, modellers still need to decide which alternative parameters should be included in the model specification. Hence, finding the "most appropriate" choice model specification still require some knowledge and intuitive judgements from modellers and domain experts.

6 Conclusion

This paper presents a novel approach for incorporating the attention mechanism into a discrete choice model. The multi-head attention layer is used to represent systematic taste variety among individuals. Experimental results using different datasets show that the proposed AttentionChoice model not only maintains the interpretability of the discrete choice model, but also significantly improves its predictive capability compared to the existing traditional and state-of-the-art schemes. This also demonstrates that the attention mechanism can indeed help to explain the systematic taste heterogeneity in choice models. In future work, we can incorporate different architectures to capture both systematic and unobserved heterogeneity. This combination could further enhance the predictive capability of these models.

Appendix A. Swissmetro

The estimated parameters

Table 9: Estimated parameters of MNL and ResLogit

Parameter (β)	MNL		ResLogit	
	Value	Std err.	Value	Std err.
SM_ASC	0.8842	4.4875e-04	0.1021	3.0548e-04
CAR_ASC	-0.7415	6.2087e-04	0.2545	5.1436e-04
TRAIN_TT	-1.3924	3.4494e-04	-0.2934	5.5974e-04
TRAIN_CO	-1.0518	2.2985e-04	0.0636	3.6082e-04
TRAIN_HE	-0.0058	1.6968e-05	-0.0010	5.8876e-06
TRAIN_GA	2.2131	3.3349e-04	0.3363	4.9760e-04
TRAIN_AGE	2.1648	1.5196e-03	-0.0623	2.2442e-04
TRAIN_MALE	0.2123	3.8940e-04	-0.2707	4.9761e-04
TRAIN_INCOME	1.7966	1.2730e-03	-0.0589	2.6666e-04
TRAIN_PURPOSE	2.1827	1.5902e-03	-0.1655	4.0184e-04
SM_TT	-1.3468	3.7303e-04	0.0590	2.5840e-04
SM_CO	-0.8443	1.9640e-04	0.0605	1.5733e-04
SM_HE	-0.0079	1.8076e-05	0.0000	2.0028e-05
SM_SEATS	0.1395	6.6466e-05	-0.0852	3.5867e-04
SM_GA	0.7858	2.1613e-04	0.0150	3.3028e-04
SM_AGE	1.9486	1.5259e-03	-0.2165	4.6356e-04
SM_MALE	0.9711	4.1701e-04	-0.1062	3.0301e-04
SM_INCOME	1.8359	1.2838e-03	-0.0833	3.4928e-04
SM_PURPOSE	2.0488	1.5979e-03	-0.3017	6.0840e-04
CAR_TT	-1.1302	2.5616e-04	-0.0536	2.5702e-04
CAR_CO	-0.7426	1.6791e-04	-0.0399	2.1752e-04
CAR_AGE	2.1055	1.5425e-03	0.1321	2.2493e-04
CAR_MALE	1.1682	4.6303e-04	0.0403	2.0397e-04
CAR_INCOME	1.7552	1.2884e-03	0.0426	1.1360e-04
CAR_PURPOSE	2.3043	1.6207e-03	0.1769	2.8603e-04
CAR_LUGGAGE	-0.0966	2.9107e-05	0.0007	3.9032e-05

Table 10: Estimated parameters of TasteNet and AttentionChoice

Parameter (β)	TasteNet		AttentionChoice	
	Value	Std err.	Value	Std err.
SM_ASC	2.4296	5.3308e-03	2.3307	2.1290e-03
CAR_ASC	1.9140	3.8875e-03	1.8642	1.8475e-03
TRAIN_TT	-0.0780	3.5969e-04	-0.3582	4.6043e-04
TRAIN_CO	-1.2101	1.9788e-03	-1.6062	1.8571e-03
TRAIN_HE	-0.0186	2.0730e-04	-0.0129	1.3314e-05
SM_TT	-0.8034	2.1158e-03	-0.8422	7.2621e-04
SM_CO	-1.0489	1.4647e-03	-1.4831	1.6090e-03
SM_HE	-0.0057	7.2669e-05	-0.0080	2.7778e-05
SM_SEATS	0.0757	5.8443e-04	0.2936	2.4190e-04
CAR_TT	-1.3153	2.0937e-03	-1.8223	1.8867e-03
CAR_CO	-0.6517	8.5816e-04	-0.9240	7.7370e-04

The cumulative distribution of parameters

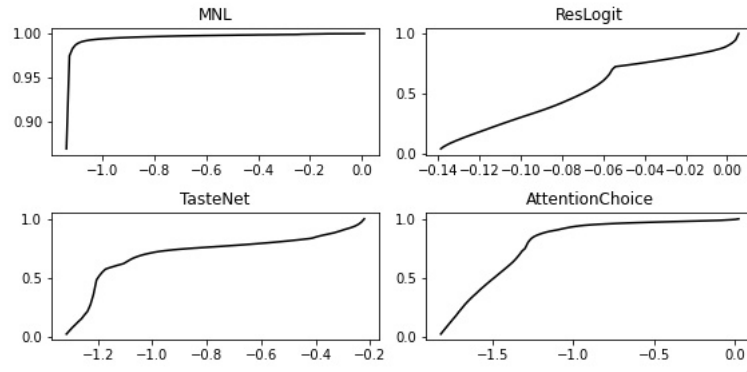


Figure 8: The cumulative distribution of β_{car_tt}

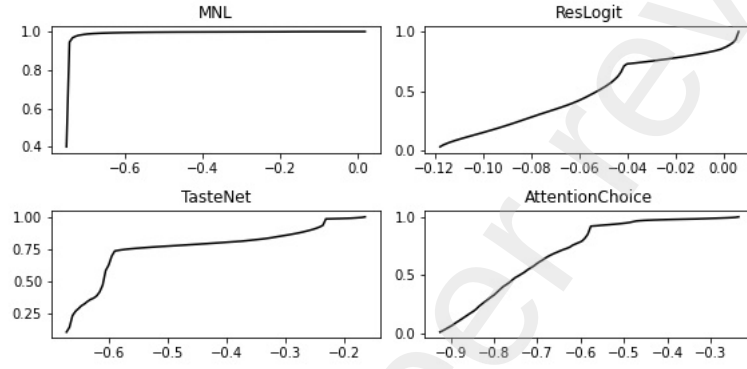


Figure 9: The cumulative distribution of β_{car_co}

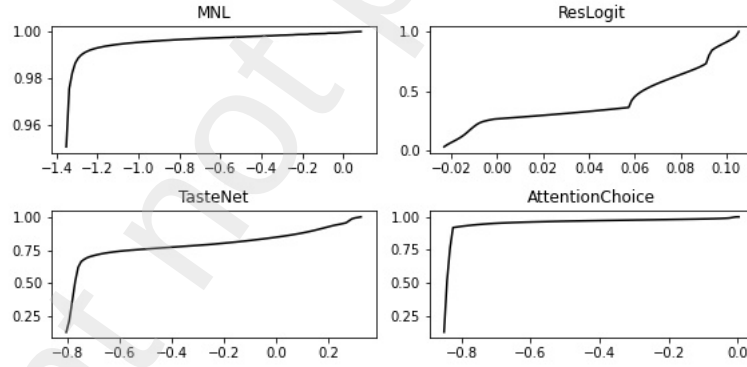


Figure 10: The cumulative distribution of β_{sm_tt}

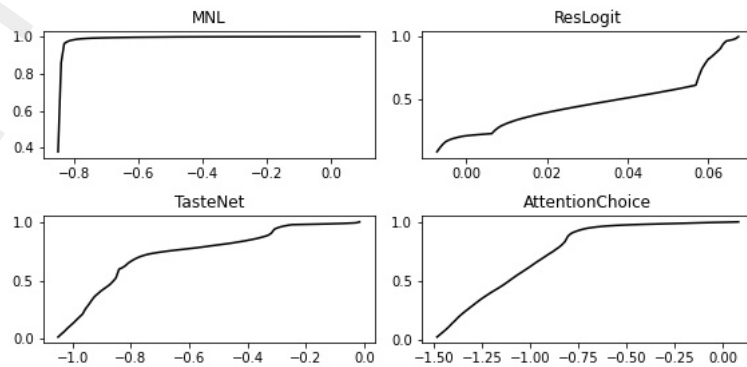


Figure 11: The cumulative distribution of β_{sm_co}

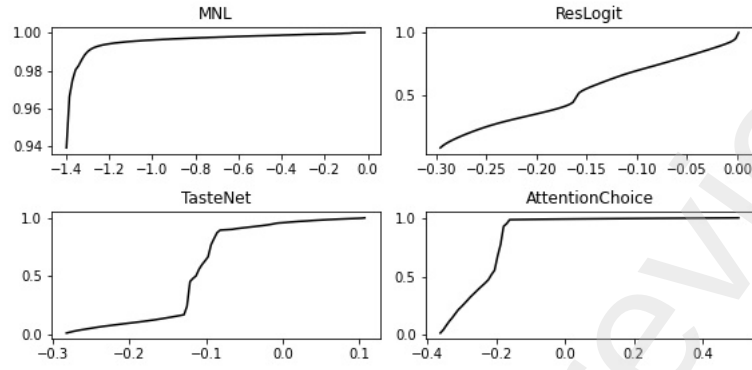


Figure 12: The cumulative distribution of β_{train_tt}

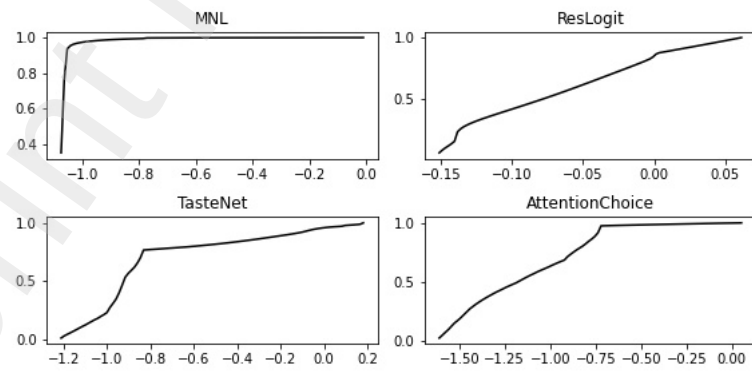


Figure 13: The cumulative distribution of β_{train_co}

Appendix B. London Passenger

The estimated parameters

Table 11: Estimated parameters of MNL and ResLogit

Parameter (β)	MNL		ResLogit	
	Value	Std err.	Value	Std err.
pt_asc	-1.1548	5.5915e-04	0.1745	3.7563e-04
cycle_asc	2.3514	1.2923e-03	0.2250	8.4550e-04
walk_asc	2.6511	1.3523e-03	-0.3815	1.0887e-03
car_purpose	-0.1014	1.1681e-04	-0.0074	2.4214e-04
car_fueltype	0.3552	3.1651e-04	-0.2186	8.0297e-04
car_start_time	-0.0778	3.3894e-05	0.0119	1.2118e-04
car_travel_month	0.0679	2.0182e-05	-0.0268	5.9178e-05
car_travel_date	0.0096	1.5105e-05	-0.0160	5.5935e-05
car_day_of_week	0.0011	2.9480e-05	0.0237	6.1747e-05
car_age	0.2355	1.0467e-04	-0.0818	2.6081e-04
car_female	0.0103	3.4943e-05	0.0150	1.7695e-04
car_driving_license	-0.1824	4.8526e-05	0.2021	6.1556e-04
car_ownership	-0.4686	2.1035e-04	0.2531	8.9354e-04
dur_driving	-0.2281	8.6613e-04	-0.2438	4.3001e-04
cost_driving	0.4986	2.3575e-04	-0.2227	5.3775e-04
pt_purpose	-0.5140	1.5713e-04	0.0162	5.7099e-04
pt_faretype	0.4251	2.9029e-04	-0.0124	4.3522e-04
pt_start_time	0.0151	2.2933e-05	-0.0251	2.7084e-04
pt_travel_month	0.1670	5.1431e-05	-0.0407	2.4801e-04
pt_travel_date	-0.0583	2.3282e-05	0.0289	2.0526e-04
pt_day_of_week	-0.0573	2.5519e-05	-0.0469	3.7901e-04
pt_interchanges	-0.6827	3.9349e-04	-0.0620	8.9778e-04
pt_age	0.1073	5.8350e-05	0.0424	5.8776e-04
pt_female	-0.5465	1.8062e-04	0.2676	1.1111e-03
dur_pt	1.9281	1.1691e-03	-0.4184	1.5479e-03
cost_transit	1.5873	8.4524e-04	0.0737	9.9207e-04
cycle_purpose	-0.2604	8.1244e-05	-0.1603	5.9403e-04
cycle_start_time	-0.0400	3.2161e-05	-0.0022	1.7434e-04
cycle_travel_month	0.1091	2.3446e-05	0.0308	1.0402e-04
cycle_travel_date	0.0378	1.5157e-05	-0.0371	1.8834e-04
cycle_day_of_week	-0.0939	2.8020e-05	-0.0074	2.2393e-04
cycle_age	0.1850	1.0848e-04	-0.0964	3.7443e-04
cycle_female	0.0954	5.0101e-05	-0.3005	1.3759e-03
dur_cycling	4.4284	2.4898e-03	0.6370	2.0410e-03
walk_purpose	-0.0446	9.1789e-05	-0.2117	1.2838e-03
walk_start_time	0.0459	2.6348e-05	0.0015	1.2566e-04
walk_travel_month	0.0580	2.5698e-05	0.0519	3.2113e-04
walk_travel_date	0.0031	1.6260e-05	-0.0732	2.5507e-04
walk_day_of_week	0.0971	2.6097e-05	0.0220	1.0480e-04
walk_Age	0.3363	9.2007e-05	-0.2024	8.3066e-04
walk_female	0.0239	3.6058e-05	-0.5681	2.0699e-03
dur_walking	3.9669	2.3788e-03	-0.6474	1.6174e-03

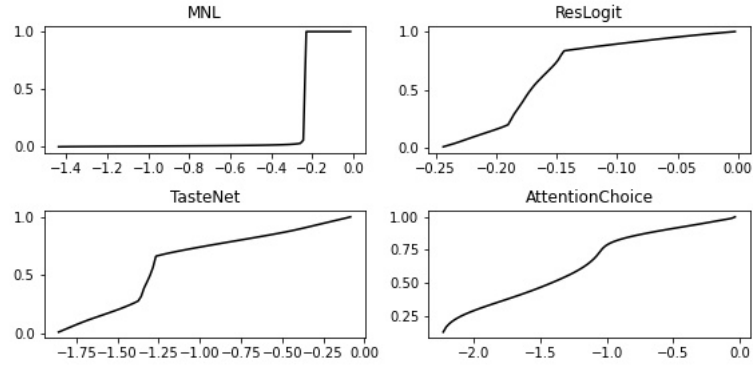


Figure 14: The cumulative distribution of $\beta_{dur_driving}$

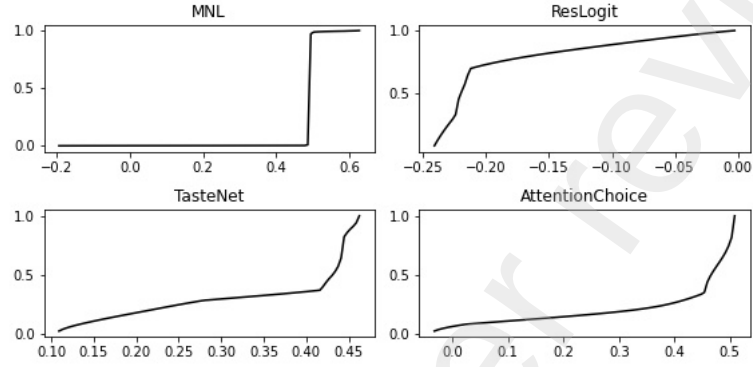


Figure 15: The cumulative distribution of $\beta_{cost_driving}$

Table 12: Estimated parameters of TasteNet and AttentionChoice

Parameter (β)	TasteNet		AttentionChoice	
	Value	Std err.	Value	Std err.
pt_asc	-0.2130	0.0031	0.2414	0.0050
cycle_asc	3.0370	0.0080	3.4248	0.0101
walk_asc	3.2505	0.0084	3.5863	0.0098
cost_driving	0.4199	0.0011	0.4592	0.0015
dur_driving	-1.8578	0.0041	-2.2289	0.0057
cost_transit	0.6392	0.0019	0.9780	0.0020
dur_pt	1.5224	0.0053	1.4192	0.0054
pt_interchanges	-0.1504	0.0010	-0.3258	0.0007
dur_cycling	4.0328	0.0119	3.9654	0.0120
dur_walking	3.4416	0.0113	3.3827	0.0111

The cumulative distribution of parameters

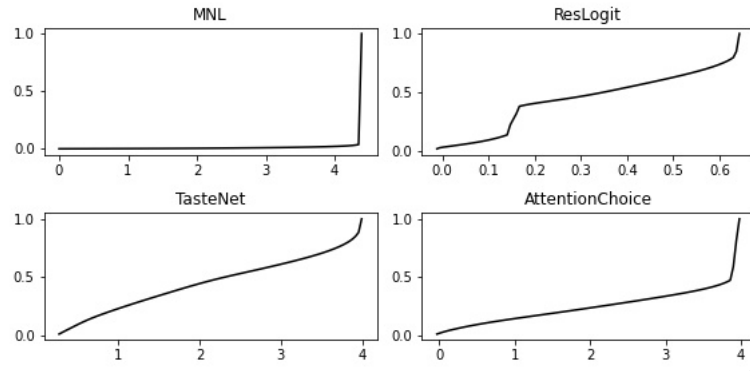


Figure 16: The cumulative distribution of $\beta_{dur_cycling}$

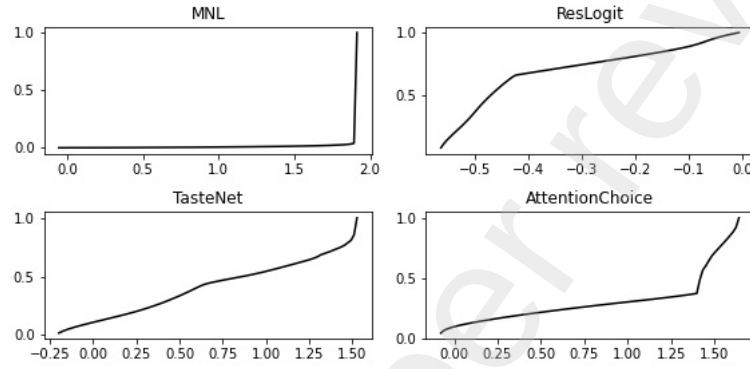


Figure 17: The cumulative distribution of β_{dur_pt}

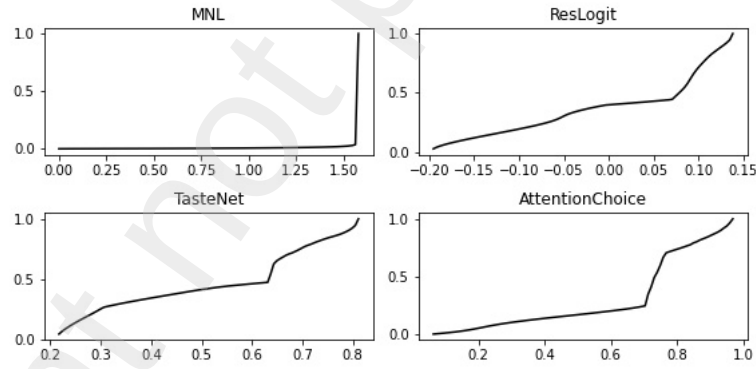


Figure 18: The cumulative distribution of $\beta_{cost_transit}$

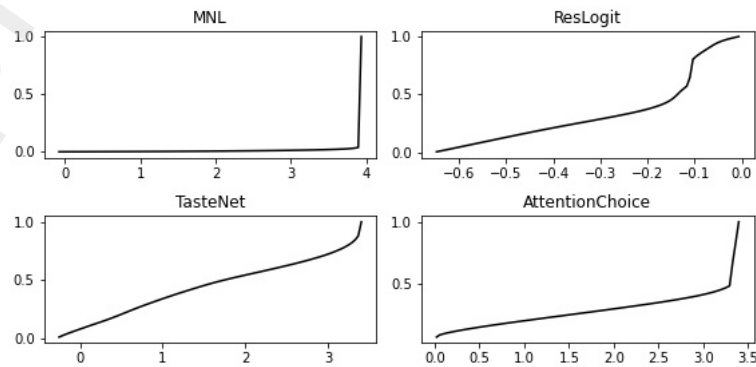


Figure 19: The cumulative distribution of $\beta_{dur_walking}$

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